

BAIO Middle Stage Curriculum (Grades 6-8)

Bharat AI Olympiad - CBSE CTAI 2026-27 Aligned Framework

1. EXAM PATTERN & MARKING SCHEME

Total Questions: 45 | Duration: 60 minutes | Total Marks: 50 Marks

Section A: CT & Logical Reasoning (15 Questions, 15 Marks)

Section B: AI & Technology Concepts (15 Questions, 15 Marks)

Section C: Everyday AI Around Us (10 Questions, 10 Marks)

Section D: Innovation Arena (5 Questions, 10 Marks - 2 Marks each)

There is no negative marking.

2. GRADE 6 SYLLABUS & OUTCOMES

Logical Units: Greedy algorithms, bubble sort full trace, binary search efficiency, Euler's formula, cyclic patterns, De Morgan's laws, algorithm efficiency O-notation introduction.

AI Units: Human vs machine intelligence, learning types in depth, four data types, digital footprints, secure passwords, phishing recognition, privacy measures.

Sample Question: A town's AI traffic system reduces total waiting time by 25% but increases waiting time at one low-income crossing by 40%. Is this fair? What design principle should be added?

3. GRADE 7 SYLLABUS & OUTCOMES

Logical Units: Recursion (base case, recursive case), divide-and-conquer (merge sort), stacks (LIFO), queues (FIFO), BFS vs DFS, XOR and De Morgan's Laws, complexity comparison.

AI Units: Regression, classification, clustering, computer vision (CNNs), NLP limitations, data science (collecting, cleaning, visualising), digital citizenship.

Sample Question: An AI loan system trained on 2010-2020 data denies women at twice the rate of equally qualified men. Give two causes of this bias and how to fix it.

4. GRADE 8 SYLLABUS & OUTCOMES

Logical Units: Dijkstra's shortest path algorithm, dynamic programming, hash tables, binary-to-decimal ($1011=11$), hexadecimal ($\#FF0000=\text{red}$), adjacency matrices.

AI Units: CTAI AI lifecycle (Define, Collect, Test, Reflect), Teachable Machine, ML for Kids, data fairness, explainable AI, DPDP Act 2023, IndiaAI Mission.

Sample Question: Design an AI project using Teachable Machine to help visually impaired students navigate school - applying all four stages of the CTAI project lifecycle.